

Name:

Class:

Date:

Python ↔ AP Pseudocode

Cheat Sheet · keep this in your notebook



Why this matters

You write Python in labs, but the AP exam uses AP pseudocode (on the Exam Reference Sheet you get during the test). Same ideas, different spelling.

Python	AP pseudocode
Output & input	
<code>print(x)</code>	<code>DISPLAY(x)</code>
<code>name = input()</code>	<code>name ← INPUT()</code>
<code># comment</code>	(comments aren't in AP pseudocode)
Variables & math	
<code>x = 5</code>	<code>x ← 5</code>
<code>x = x + 1</code>	<code>x ← x + 1</code>
<code>a + b, a - b, a * b, a / b</code>	<code>a + b, a - b, a * b, a / b</code>
<code>a % b</code>	<code>a MOD b</code>
Comparisons & logic	
<code>a = b</code>	<code>a = b</code>
<code>a ≠ b</code>	<code>a ≠ b</code>
<code>a ≥ b, a ≤ b</code>	<code>a ≥ b, a ≤ b</code>
<code>and, or, not</code>	<code>AND, OR, NOT</code>
Selection	
<pre>if x > 5: ... else: ...</pre>	<pre>IF (x > 5) { ... } ELSE { ... }</pre>
<pre>if a: ... elif b: ...</pre>	<pre>IF (a) { ... } ELSE { IF (b) { ... } }</pre>
Iteration	
<code>for i in range(n):</code>	<code>REPEAT n TIMES</code>
<code>while x < 10:</code>	<code>REPEAT UNTIL (x ≥ 10) ← flip the condition!</code>
<code>for item in nums:</code>	<code>FOR EACH item IN nums</code>

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Cheat Sheet (continued)

Python	AP pseudocode
Lists	
nums = [4, 7, 9]	nums ← [4, 7, 9]
nums[0] (first)	nums[1] (first)
nums[-1] / nums[len(nums)-1]	nums[LENGTH(nums)]
nums.append(5)	APPEND(nums, 5)
nums.insert(0, 2)	INSERT(nums, 1, 2)
nums.pop(0)	REMOVE(nums, 1)
len(nums)	LENGTH(nums)
[]	[]
Procedures	
def area(w, h): return w * h	PROCEDURE area(w, h) { RETURN (w * h) }
a = area(3, 4)	a ← area(3, 4)
Random & strings	
random.randint(1, 6)	RANDOM(1, 6) (both include 1 and 6)
s + t	concat(s, t) (defined in the question)
s[0:3]	substring(s, 1, 3) (defined in the question)

Top 10 Traps

1. AP lists start at index **1**. Python starts at 0.
2. ← assigns; = compares. Python's == is AP's =.
3. **REPEAT UNTIL** stops when the condition is TRUE — the opposite of while.
4. **RETURN** ends the procedure immediately.
5. **RANDOM(a, b)** includes both a and b: RANDOM(1, 6) has 6 values.
6. Index below 1 or above LENGTH → **error**; the program stops.
7. INSERT shifts later items **right**; REMOVE shifts them **left** — rewrite the list after every step.
8. On the AP reference sheet, b ← a gives b a **copy** of list a.
9. MOD has the same precedence as * and /. Work left to right.
10. String procedures like substring are **defined in each question**. Read the definition!